Exercise Sheet 12

Discrete Mathematics by Hongfei Fu, 2023.11.20

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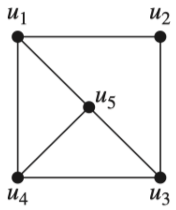
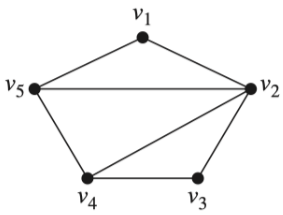
1. ([R], Page 676-677, Exercise 36, 37, 42, 43(**2pt**), 44(**2pt**)) In Exercises 36, 37, 42, 43, 44 determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.

(36).



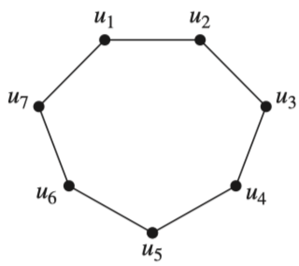


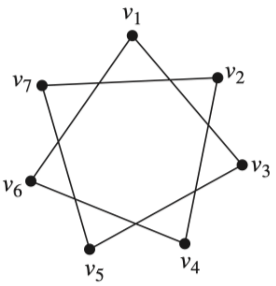
Answer Area:

No. A proof is as follows: $\because$ In the left graph, there isn't a vertex whose degree is 4. But in the right graph, $\deg(v_2) = 4$.

$\therefore$ The two graphs are not isomorphic.

(37).





Answer Area:

Yes. There is a bijection:

$f(u_1) = v_1$

$f(u_2) = v_3$

$f(u_3) = v_5$

$f(u_4) = v_7$

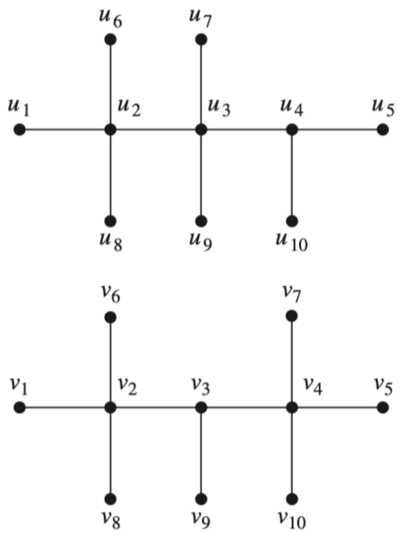
$f(u_5) = v_2$

$f(u_6) = v_4$

$f(u_7) = v_6$

Hence, the two graphs are isomorphic.

(42).



Answer Area:

No. A proof is as follows: Assuming that there is a bijection that satisfies the needs.

$\because$If $f(u_i) = v_j$, then $deg(u_i) = deg(v_j)$

$\therefore f(u_5) = v_5$ or $f(u_5) = v_1$.

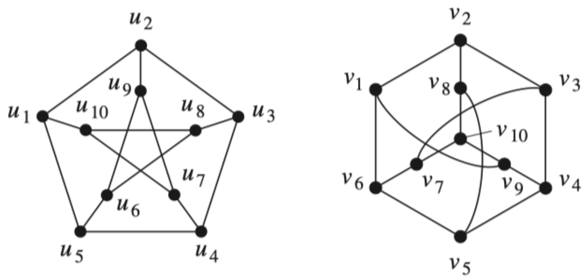
$\because deg(v_2) = deg(v_4) = 4 \neg 3 = deg(u_4)$.

$\therefore f(u_2) \neg v_4$ and $f(u_2) \neg v_2$.

$\therefore f(u_5) \neg v_5$ and $f(u_5) \neg v_1$

Contradict. The two graphs are not isomorphic.

(43)(**2pt**).



Answer Area:

Yes. A bijectionYes. There is a bijection:

$f(u_1) = v_1$

$f(u_2) = v_2$

$f(u_3) = v_3$

$f(u_4) = v_7$

$f(u_5) = v_6$

$f(u_6) = v_5$

$f(u_7) = v_{10}$

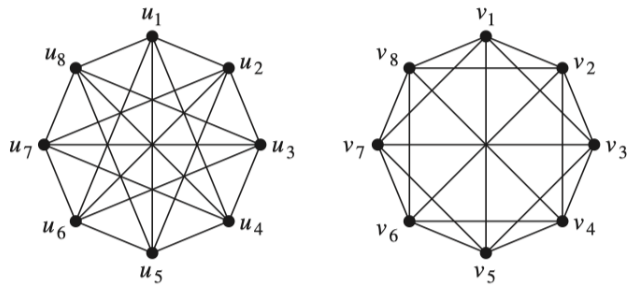
$f(u_8) = v_4$

$f(u_9) = v_8$

$f(u_{10}) = v_9$

Hence, the two graphs are isomorphic.

(44)(**2pt**).



Answer Area:

No. A proof is as below:

Let $\mathbf{G_1} = (\mathbf{V_1}, \mathbf{E_1}), \mathbf{G_2} = (\mathbf{V_2}, \mathbf{E_2})$ representing two graphs. Assuming that there is a bijection $f : \mathbf{V_1} \to \mathbf{V_2}$ such that $\forall u, v \in \mathbf{V_1}(u, v \in \mathbf{E_1} \leftrightarrow f(u), f(v) \in \mathbf{E_2})$.

According to the symmetry, we can define $f(u_1) = v_1$.

Also according to the symmetry, we can conclude that $f(u_5) = v_5$.

However, all the neighbours of u_5 are the neighbours of u_1 and some of the neighbours of v_5 aren't the neighbours of v_1.

Therefore, contradict.

So there isn't such bijection. The two graphs are not isomorphic.